

DOMES CONSTRUCTION DATA SHEET

Material Costs, Technical Notes, Work Plans & Crew Requirements

Compiled: September 2026 | Currency: USD | Note: Prices are indicative for US/EU markets; regional variations ±30-50%

TABLE 1 — MATERIAL PRICE COMPARISON (Per m² of Dome Shell)

Material	Price/m² (Materials)	Price/m² (Installed)	Typical Thickness	Dome Suitability
Shotcrete (Wet-Mix)	\$35 – \$75	\$65 – \$215	7.5 – 15 cm	★★★★★ Excellent
Shotcrete (Dry-Mix / Guniting)	\$30 – \$65	\$60 – \$180	7.5 – 15 cm	★★★★★ Excellent
Compressed Earth Block (CEB)	\$15 – \$45	\$80 – \$200	25 – 35 cm	★★★☆☆ Moderate*
Masonry (Brick/Stone Dome)	\$45 – \$120	\$150 – \$400	20 – 45 cm	★★★★☆ Good**
Concrete (Monolithic/Poured)	\$50 – \$100	\$430 – \$860 (shell only)	7.5 – 20 cm	★★★★★ Excellent
Brick (Traditional)	\$32 – \$86	\$110 – \$325	11.5 – 23 cm	★★★☆☆ Moderate*

* CEB and brick are rarely used for true self-supporting hemispherical domes without reinforcement; better for stem walls or Nubian vaults.

** Masonry domes require extensive centering and master craftsmen; practical limit ~10-15 m diameter.

TABLE 2 — SHOTCRETE (WET-MIX & DRY-MIX / GUNITE)

A. Technical Notes

Parameter	Wet-Mix Shotcrete	Dry-Mix (Gunite)
Mix Delivery	Pre-mixed concrete delivered via pump	Dry mix via hose; water added at nozzle
Rebound Loss	5 – 15%	15 – 30%
Application Speed	Faster for large volumes	Slower; better for detail work
Dust Level	Lower	Higher
Surface Prep	Clean, DAMP surface required	Clean, DRY surface required
Thickness Control	Easier to control	Requires highly skilled nozzleman
Curing	Standard wet curing or compounds	More prone to shrinkage; wet curing essential
Best for Domes	Large structural domes, monolithic shells	Small repairs, decorative curves, pools

B. Preparation Required Before Application

- 1. Surface Cleaning** — Remove all loose material, debris, oil, and contaminants. Surface must be structurally sound.
- 2. Surface Conditioning** — Wet-mix: dampen surface (SSD — Saturated Surface Dry). Dry-mix: ensure surface is dry.
- 3. Rebar / Mesh Installation** — Install engineered rebar pattern (hoop + vertical bars) or welded wire mesh. Rebar hangers embedded in foam (for monolithic domes).
- 4. Formwork / Airform** — For monolithic domes: inflate PVC airform membrane; apply polyurethane foam first; embed rebar hangers.
- 5. Equipment Setup** — Position concrete pump, run hoses, test air compressor (minimum 250–350 cfm), check nozzle operation.
- 6. Mock-up Panel** — Spray a test panel to verify mix design, nozzleman skill, and rebound rate.

C. Steps of Work (per 100 m²)

Step	Activity	Duration
1	Surface prep, cleaning, rebar inspection	4 – 6 hours
2	Equipment mobilization, hose layout, test spray	2 – 3 hours
3	First layer application (scratch coat, 1–2 cm)	2 – 3 hours
4	Rebar embedding check, surface roughening	1 – 2 hours
5	Second layer application (to design thickness)	3 – 4 hours
6	Trim, finish, thickness verification	2 – 3 hours
7	Curing initiation (wet burlap / misting / compounds)	Immediate; 7–14 days
8	Equipment cleanup, rebound removal	2 – 3 hours

D. Team Required (Total: 6 – 9 persons per shift)

Role	Count	Responsibilities
Nozzleman	1	Controls spray pattern, thickness, water addition (dry-mix)
Pump Operator	1	Manages concrete pump, flow rate, hose pressure
Nozzleman Helper	1 – 2	Operates air blowpipe to remove rebound, moves hoses
Rebar Crew	2 – 3	Installs/hangs rebar, checks spacing, ties wire
Curing Monitor	1	Ensures continuous moisture during curing period

E. Materials Required (per 100 m² at 7.5 cm thickness)

Material	Quantity	Notes
Shotcrete concrete	7.5 m ³	Add 10–15% waste factor for rebound
Rebar (varies by design)	800 – 1,500 kg	Hoop + vertical pattern; #3 to #6 bars typical
Rebar hangers / chairs	200 – 400 pcs	Embedded in substrate or foam
Curing compound	20 – 30 L	Or wet burlap + plastic sheeting
Water for curing	500 – 1,000 L	Continuous misting for 7 days minimum

F. Equipment Required

- Concrete pump (boom or line pump)
- Air compressor (250–350 cfm minimum)
- Shotcrete nozzle + hoses (50–100 m)
- Rebar cutter/bender
- Spray curing equipment or burlap rolls
- Thickness gauge / depth pins

TABLE 3 — COMPRESSED EARTH BLOCK (CEB)

A. Technical Notes

Parameter	Specification
Composition	90–94% soil/sand mix + 6–10% Portland cement or lime stabilizer
Block Size (typical)	10" × 14" × 4" (25 × 35 × 10 cm) — varies by press
Block Weight	15–25 kg depending on size and moisture
Compressive Strength	3–7 MPa (stabilized); up to 10+ MPa with higher cement content
Water Resistance	Good when stabilized ($\geq 10\%$ cement); poor if unstabilized
Blocks per m ² of Wall	~11 blocks (face area) for 10" thick wall
Thermal Mass	Excellent — high thermal inertia
Dome Suitability	Best for corbelled/Nubian vaults or stem walls; true hemispherical domes require reinforcement

B. Preparation Required Before Construction

- Soil Testing** — Test local soil for clay content (ideal: 15–30% clay), silt, sand, gravel. Perform shrinkage and compression tests.
- Sieving / Crushing** — Remove organic matter, large stones. Pass through 20–25 mm screen.
- Stabilizer Mixing** — Blend cement or lime thoroughly with dry soil (minimum 5 minutes in pan mixer).
- Moisture Conditioning** — Add water to reach optimum moisture content (typically 10–14%). Mix until homogeneous.
- Block Production** — Press blocks using mechanical press (manual or hydraulic). Cure for 7–28 days under plastic sheeting.
- Foundation Prep** — Pour reinforced concrete ring beam or stem wall. Ensure level and plumb.
- Centering (for domes)** — Erect temporary wooden/metal formwork if building a true dome profile.

C. Steps of Work (per 100 m² wall)

Step	Activity	Duration
1	Soil excavation, testing, sieving	1 – 2 days
2	Mixing soil + stabilizer + water	4 – 6 hours
3	Block pressing, stacking, initial curing	2 – 3 days (ongoing)
4	Block curing (7–14 days minimum)	7 – 14 days
5	Foundation / ring beam construction	2 – 3 days
6	First course laying, leveling	4 – 6 hours
7	Subsequent courses with mortar	2 – 3 days
8	Surface finishing (clay plaster / stucco / sealer)	2 – 3 days

D. Team Required (Total: 8 – 13 persons)

Role	Count	Responsibilities
Machine Operator	1	Operates CEB press, adjusts pressure and mix
Mixing Crew	2 – 3	Shovel soil, add stabilizer/water, load mixer
Block Handlers	2 – 3	Transport green blocks to curing stacks
Mason / Lead Builder	1 – 2	Lays blocks, checks level, manages bond pattern
Laborers	2 – 4	General assistance, mortar mixing, material transport

E. Materials Required (per 100 m² of 25 cm thick wall)

Material	Quantity	Notes
CEB blocks	1,100 – 1,200 pcs	10" × 14" face; includes 5% wastage
Soil / subsoil	25 – 30 tonnes	On-site sourcing preferred
Portland cement (stabilizer)	2.5 – 3.5 tonnes	10% stabilization ratio
Water	3,000 – 5,000 L	For mixing and block curing
Mortar (cement:sand)	2.5 – 3.5 m ³	1:6 ratio for laying; thin-bed possible
Plastic sheeting (curing)	150 – 200 m ²	Cover stacks for moisture retention

F. Equipment Required

- CEB press (manual or hydraulic — \$2,000–\$35,000)
- Pan mixer or drum mixer
- Soil sieve / crusher
- Wheelbarrows, shovels, buckets
- Masonry tools: levels, string lines, trowels, jointers
- Scaffolding (for dome walls > 2 m)

TABLE 4 — MASONRY DOME (Brick or Stone)

A. Technical Notes

Parameter	Specification
Structural Behavior	Compression-only structure; relies on hoop tension resistance via geometry
Laying Pattern	Herringbone, radial, or spiral courses; each course is self-supporting as it rises
Bond Type	English Bond, Flemish Bond, or radial dry-stone with mortar joints
Centering Required	Yes — temporary support until keystone / closing ring is placed
Maximum Span	Historically 40–50 m; practical modern limit ~10–15 m without reinforcement
Mortar Ratio	1:3 to 1:6 (cement:sand) for brick; lime mortar for historic/traditional builds
Dome Suitability	Excellent for traditional architecture; labor-intensive

B. Preparation Required Before Construction

- Foundation** — Pour reinforced concrete ring beam or footing sized for total dome weight + live loads.
- Centering / Formwork** — Erect temporary wooden or metal framework defining the dome's interior profile. Must be strong enough to support wet masonry + workers.
- Material Staging** — Stockpile bricks/stones, sand, cement near work area. Pre-cut any special shapes.
- Layout** — Mark radial lines from center; establish first course line on ring beam.
- Mortar Batching Setup** — Install mixer, water supply, and mortar boards at working height.
- Safety** — Install edge protection, harness points, and debris netting.

C. Steps of Work (per 100 m² dome surface)

Step	Activity	Duration
1	Foundation and ring beam construction	3 – 5 days
2	Centering erection	2 – 4 days
3	First 3–4 courses (base rings)	2 – 3 days
4	Rising courses to springing line	3 – 5 days
5	Haunch and crown courses	3 – 5 days
6	Keystone / closing ring placement	4 – 8 hours
7	Centering striking (gradual removal)	1 – 2 days
8	Pointing, joint finishing	2 – 3 days
9	Waterproofing / exterior coating	2 – 3 days

D. Team Required (Total: 9 – 16 persons)

Role	Count	Responsibilities
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Master Mason / Dome Specialist	1 – 2	Layout, pattern control, critical course placement
Journeyman Masons	2 – 4	Lay bricks/stones, mortar work
Mortar Mixers	1 – 2	Continuous mortar supply, consistency control
Material Handlers	2 – 4	Deliver bricks, stones, mortar to work face
Centering Carpenters	2 – 3	Erect, adjust, and later remove formwork
Safety Monitor	1	Harness checks, edge protection, debris management

E. Materials Required (per 100 m² of 23 cm thick dome)

Material	Quantity	Notes
Bricks (standard)	5,000 – 5,500 pcs	Includes 5% wastage; modular or traditional size
OR Natural Stone	12 – 18 tonnes	Varies by stone density and wall thickness
Cement (for mortar)	25 – 35 bags (50 kg)	1:4 to 1:6 mortar ratio
Sand	8 – 12 m ³	Clean, sharp masonry sand
Water	2,000 – 3,000 L	For mortar and curing
Centering timber / steel	30 – 50 m ² contact area	Reusable; rental or purchase
Waterproofing membrane / coating	110 – 120 m ²	Overlaps at joints

F. Equipment Required

- Masonry mixer (continuous feed preferred)
- Scaffolding / work platforms (radial or circular)
- Hoist / crane (for stone domes > 8 m diameter)
- Centering formwork system (custom-built)
- Masonry cutting saw
- Levels, plumb bobs, radius gauges, string lines
- Pointing tools, joint rakes

TABLE 5 — CONCRETE (MONOLITHIC DOME — POURED / SPRAYED)

A. Technical Notes

Parameter	Specification
Construction Method	Airform-inflated membrane; foam + rebar + shotcrete applied internally
Shell Composition	Airform (roof membrane) + PU foam (insulation) + rebar + shotcrete (structure)
Typical Thickness	7.5 cm (3 in) small domes; up to 20 cm for large / load-bearing structures
Rebar Pattern	Engineered hoop (horizontal) + vertical bars; double mat for large domes
Foam Thickness	7.5–10 cm (3–4 in) typical; provides R-30 to R-60 insulation
Compressive Strength	3,000–5,000 PSI shotcrete
Total System Cost	\$150 – \$250/sq ft (\$1,614 – \$2,691/m ²) finished
Shell-Only Cost	\$40 – \$80/sq ft (\$430 – \$860/m ²) for shotcrete + rebar + foam + airform
Foundation	\$6 – \$14/sq ft (\$65 – \$151/m ²) for ring beam or slab-on-grade
Dome Suitability	The modern standard for disaster-resistant domes

B. Preparation Required Before Construction

- 1. Site Prep** — Clear, grade, and compact building pad. Install drainage.
- 2. Foundation / Ring Beam** — Pour reinforced concrete ring beam with vertical rebar dowels extending upward.
- 3. Anchor Bolt Layout** — Install anchor bolts for airform attachment at precise spacing.
- 4. Airform Manufacturing** — Custom PVC-coated fabric membrane manufactured to exact dome dimensions.
- 5. Airform Installation** — Attach to ring beam, seal all seams, install inflation blower fans.
- 6. Inflation** — Pressurize airform to design pressure (typically 1.5–2.0 in water column). Maintain throughout construction.
- 7. Airlock Entry** — Install double-door airlock system for worker access without pressure loss.
- 8. Material Staging** — Position foam rigs, shotcrete pump, rebar supplies inside the airform.

C. Steps of Work

Step	Activity	Duration
1	Foundation excavation, forms, rebar, pour	5 – 10 days
2	Airform delivery, attachment, inflation	2 – 3 days
3	Polyurethane foam application (interior)	3 – 5 days
4	Rebar hanger embedment in foam	Concurrent with foam
5	Rebar installation (hoop + vertical)	5 – 10 days
6	Shotcrete application (interior, overhead)	5 – 10 days
7	Concrete curing (7–14 days minimum)	7 – 14 days
8	Blower shutdown after concrete set	Day 14–21

9	Window/door cutouts, augment frames	3 – 5 days
10	Interior finishing, utilities	30 – 60 days

D. Team Required (Total: 12 – 20 persons during active shell construction)

Role	Count	Responsibilities
Concrete Foreman	1	Overall coordination, quality control
Foam Applicators	2 – 3	Spray closed-cell PU foam to specified thickness
Rebar Crew	3 – 5	Hang and tie rebar per engineered pattern
Shotcrete Nozzleman	1 – 2	Spray concrete overhead in layers
Pump Operator	1	Manage concrete pump and hose layout
Nozzleman Helper	2	Rebound removal, hose handling, thickness checks
Airform / Pressure Monitor	1	Monitor inflation pressure 24/7 during construction
Curing Specialist	1	Ensure proper moisture retention during set

E. Materials Required (per 100 m² of Dome Surface, 7.5 cm Shell)

Material	Quantity	Notes
Airform membrane	120 – 130 m ²	Custom PVC-coated fabric; includes overlaps
Closed-cell polyurethane foam	7.5 – 10 m ³	R-value ~6 per inch; 3–4 in typical
Rebar (#3 to #5 typical)	1,200 – 2,000 kg	Hoop + vertical; spacing per engineering
Rebar hangers / chairs	300 – 500 pcs	Embedded in foam layer
Shotcrete / concrete	7.5 – 10 m ³	Add 10% waste; 3,000+ PSI
Curing compound / water	As needed	Wet curing critical for thin shell
Ring beam concrete	3 – 5 m ³	Reinforced with vertical dowels
Ring beam rebar	200 – 400 kg	#6 to #8 bars typical

F. Equipment Required

- Airform inflation blowers (2–3 units for redundancy)
- PU foam spray rig (high-pressure, heated)
- Concrete pump + 100+ m of hose
- Air compressor (350+ cfm)
- Rebar cutter/bender
- Concrete vibrator (for ring beam)
- Scaffold / work platform (circular, inside airform)
- Pressure gauges, backup power for blowers

TABLE 6 — TRADITIONAL BRICK (For Dome Base Walls or Small Domes)

A. Technical Notes

Parameter	Specification
Brick Type	Common clay, fly ash, or fire brick (for high-heat applications)
Wall Thickness	11.5 cm (4.5 in) half-brick; 23 cm (9 in) full-brick; 35 cm+ for load-bearing
Mortar Ratio	1:6 (cement:sand) standard; 1:4 for high-strength or damp conditions
Bricks per m ²	~49–60 bricks (standard 230 × 110 × 75 mm with 10 mm joints)
Compressive Strength	5–20 MPa depending on brick class
Bond Pattern	Stretcher bond (half-brick); English bond (full-brick)
Dome Suitability	Used for stem walls, base structures, or small corbelled domes

B. Preparation Required Before Construction

- Foundation** — Pour reinforced concrete footing or plinth beam; ensure level within 5 mm.
- Damp-Proof Course (DPC)** — Install DPC at plinth level to prevent rising damp.
- Material Staging** — Stack bricks near work area; protect from rain. Stage sand, cement, water.
- Layout** — Mark wall lines, corners, opening positions. Set up profiles / string lines.
- Mortar Setup** — Position mixer, measure boxes for batching, prepare mortar boards.
- Soaking Bricks** — Immerse bricks in water for 2+ hours before laying (prevents suction).

C. Steps of Work (per 100 m² of 23 cm wall)

Step	Activity	Duration
1	Foundation / footing construction	3 – 5 days
2	DPC installation, first course layout	4 – 6 hours
3	Brick soaking (2+ hours before laying)	Ongoing
4	Course-by-course laying, level/plumb checks	4 – 6 days
5	Lintel / ring beam over openings	1 – 2 days
6	Curing (water spray 2× daily)	7 days
7	Pointing / joint finishing	2 – 3 days
8	Plastering / finishing (if required)	3 – 5 days

D. Team Required (Total: 7 – 10 persons)

Role	Count	Responsibilities
Mason (1st Class)	2 – 3	Lays bricks, checks level, plumb, and alignment

Mason Helper / Mazdoor	3 – 4	Mortar transport, brick soaking, material supply
Mortar Mixer	1	Batch mortar, maintain consistency
Scaffolding Crew	1 – 2	Erect/move scaffolding as wall rises

E. Materials Required (per 100 m² of 23 cm thick wall)

Material	Quantity	Notes
Bricks (first class, 230×110×75 mm)	5,000 – 5,500 pcs	Includes 5% wastage
Cement (50 kg bags)	25 – 30 bags	1:6 mortar ratio
Sand (clean, sharp)	8 – 10 m ³	Masonry sand
Water	2,000 – 3,000 L	For mortar, soaking, and curing
Scaffolding	100 – 150 m ² rental	Per level as wall rises
DPC membrane	10 – 15 Lm	At plinth level

F. Equipment Required

- Masonry mixer (0.5–1 m³ capacity)
- Wheelbarrows, mortar pans, hods
- Scaffolding system (cup-lock or frame)
- Brick soaking tank / tarp-lined area
- Levels, plumb bobs, line pins, story poles
- Brick hammer, bolster chisel, jointing tool
- Water tank / hose for curing

TABLE 7 — QUICK REFERENCE: PROS & CONS FOR DOME CONSTRUCTION

Material	Pros	Cons
Shotcrete	Simple application; excellent bond to rebar; monolithic shell; no formwork needed after stir (5-80%)	Skilled nozzleman; equipment-intensive
CEB	Low material cost; if soil on-site; sustainable; excellent thermal insulation	Requires stabilization for load-bearing; not ideal for true domes without reinforcement
Masonry/Brick	Time-tested; uses local materials; proven centuries	Extremely labor-intensive; no steel reinforcement; plaster/cast design; centering/formwork is major cost; limited span;
Polyester Concrete	Fastest to construct; strongest disaster resistance; lightweight	Highly flammable; specialized equipment; only for one-time use; single-use membrane; requires continuous power
Brick (Traditional)	Widely available; familiar trades; low cost	Not self-supporting in true dome geometry without reinforcement; mortar joints are weak points; thermal

TABLE 8 — ESTIMATED TOTAL DOME COSTS BY SIZE (Monolithic / Shotcrete System)

Dome Diameter	Floor Area	Shell-Only Cost	Total Finished Cost
10 m (33 ft)	80 m²	\$35,000 – \$70,000	\$130,000 – \$215,000
12 m (39 ft)	115 m²	\$50,000 – \$100,000	\$185,000 – \$310,000
15 m (49 ft)	180 m²	\$75,000 – \$155,000	\$290,000 – \$485,000
20 m (66 ft)	315 m²	\$135,000 – \$270,000	\$510,000 – \$850,000
30 m (98 ft)	710 m²	\$305,000 – \$610,000	\$1,150,000 – \$1,900,000

Shell-only includes: foundation ring beam, airform, foam, rebar, shotcrete. Total finished includes: shell + windows, doors, interior walls, MEP, finishes.

FOOTNOTES & ASSUMPTIONS

- Prices are indicative for US/EU markets as of 2026. Regional variations can be ±30–50%.
- Labor rates assume \$45–\$85/hour for skilled trades (US average). Adjust for local markets.
- Dome geometry affects material quantities significantly. A hemisphere uses ~2× the surface area of its floor plan.
- Reinforcement quantities vary by seismic zone, wind load, and local building codes. Always consult a structural engineer.
- Waste factors are included in material quantities. Order 5–10% extra for contingencies.
- Curing is critical for all cement-based materials. Inadequate curing reduces strength by 30–50%.
- Shotcrete prices for dome shells are typically quoted as part of a monolithic dome package. Standalone shotcrete may differ.
- CEB costs drop dramatically with on-site soil and volunteer labor (as low as \$5–\$10/m² in self-build contexts).
- Masonry dome construction requires experienced craftsmen; training costs should be factored if not locally available.

10. Monolithic dome construction requires a licensed Monolithic Dome Institute builder or equivalent for warranty and insurance.

SOURCES

ACI 506R Guide to Shotcrete | Monolithic Dome Institute | One Community Global | Angi Construction Data | Checkatrade UK | Yojo Construction Analytics | RSMeans Construction Cost Data | Industry standard references